Super Store

Optimalisasi Profit dengan Machine Learning

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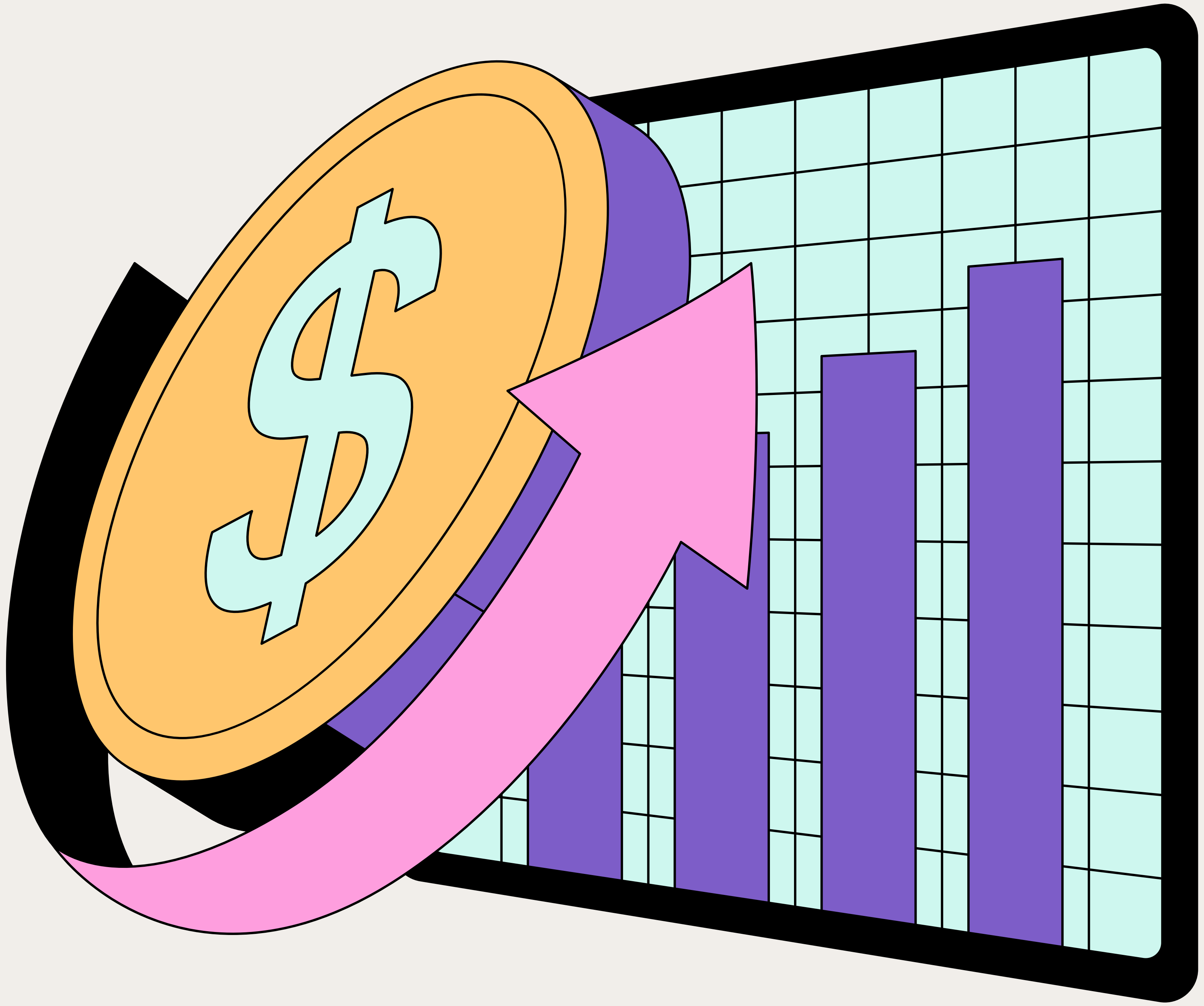


Tujuan

Memprediksi Profit dan Rugi Order Berdasarkan Pola Historis untuk Optimasi Keputusan Bisnis

Mengetahui variabel yang paling berpengaruh terhadap profit

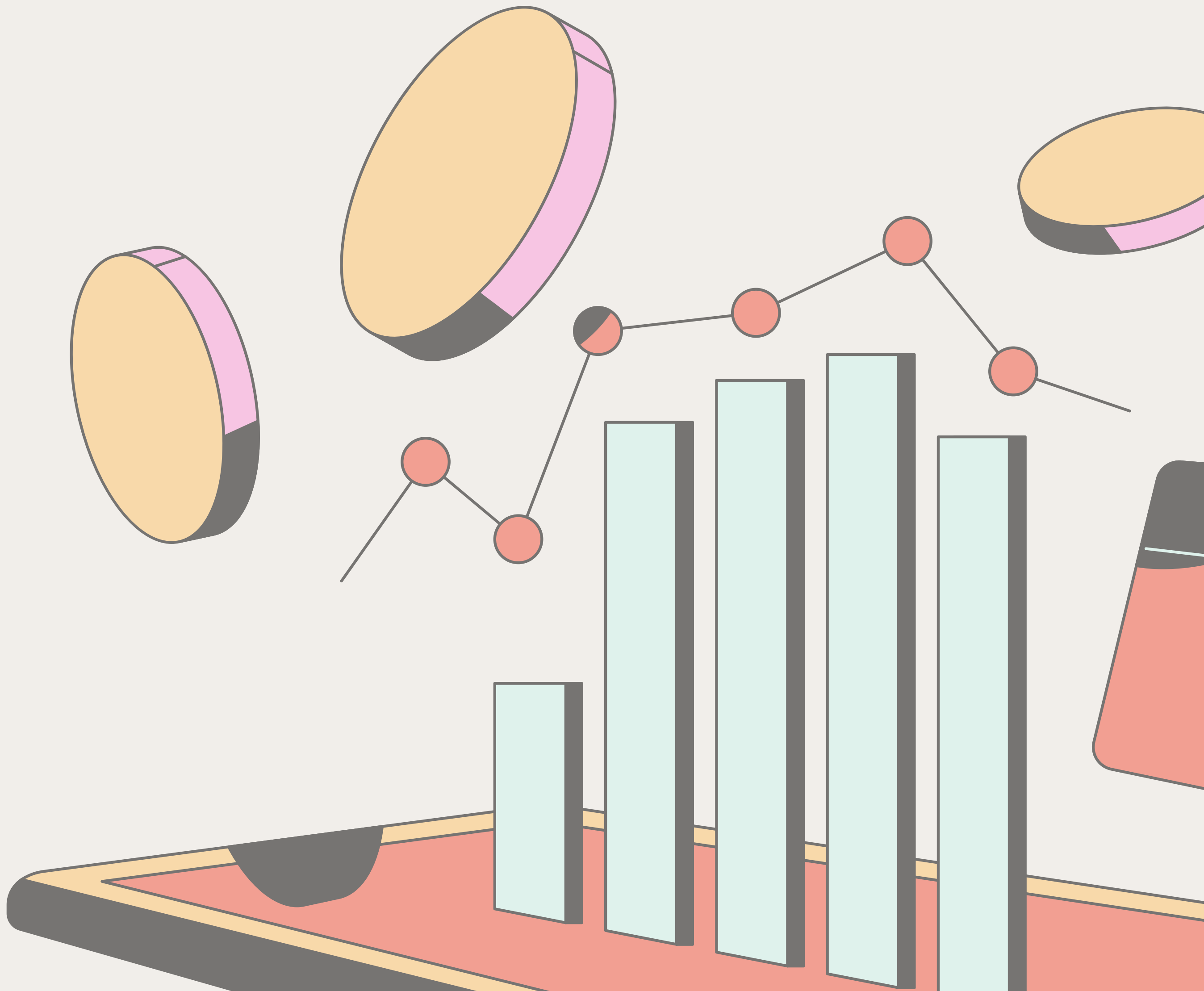
Dataset

FEATURE

- DISCOUNT
- DISCOUNT_BIN
- SALES
- CATEGORY
- MARKET
- SUB_CATEGORY
- ORDER_PRIORITY
- SEGMENT
- SHIPPING_COST
- PRODUCT_NAME
- COUNTRY
- REGION
- QUANTITY
- SHIP_DATE
- CUSTOMER_NAME
- SHIP_MODE

TARGET

- PROFIT_BIN



Profit dan Discount Bin

Kelas	Syarat	Penjelasan
`'Rugi'`	`p < 0`	Jika profit **negatif (rugi)**
`'Untung Kecil'`	`0 ≤ p ≤ 30`	Jika profit **sedikit**
`'Untung Besar'`	`p > 30`	Jika profit **besar**

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=== Statistik Profit ===
Mean   : 28.64
Median : 9.24
Modus  : [0.0]
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`discount_bin`	Range Diskon (discount)	Penjelasan
0	0.00	**Tanpa diskon**
1	>0.00 s.d. 0.10	**Diskon rendah** (0–10%)
2	>0.10 s.d. 0.30	**Diskon sedang** (10–30%)
3	>0.30 s.d. max_val (mungkin 0.8–1.0)	**Diskon tinggi** (>30%)

Feature Importance by Chi Squared

=========== Hasil Uji Chi-Squared terhadap 'profit_bin' ===========

Feature	Chi2 Hitung	Chi2 Tabel	p-value	Kesimpulan
discount_bin	36065.6884	12.5916	0.0000	✅ Tolak H0
product_name	28422.4752	7777.5746	0.0000	✅ Tolak H0
country	24149.4983	332.8538	0.0000	✅ Tolak H0
sub_category	10604.9665	46.1943	0.0000	✅ Tolak H0
category	6220.4128	9.4877	0.0000	✅ Tolak H0
ship_date	4812.2120	3052.9553	0.0000	✅ Tolak H0
region	3323.9449	36.4150	0.0000	✅ Tolak H0
customer_name	2643.4157	1681.8204	0.0000	✅ Tolak H0
market	1397.6546	21.0261	0.0000	✅ Tolak H0
order_priority	4.2271	12.5916	0.6460	Gagal Tolak H0
ship_mode	3.1470	12.5916	0.7902	Gagal Tolak H0
segment	1.0461	9.4877	0.9027	Gagal Tolak H0

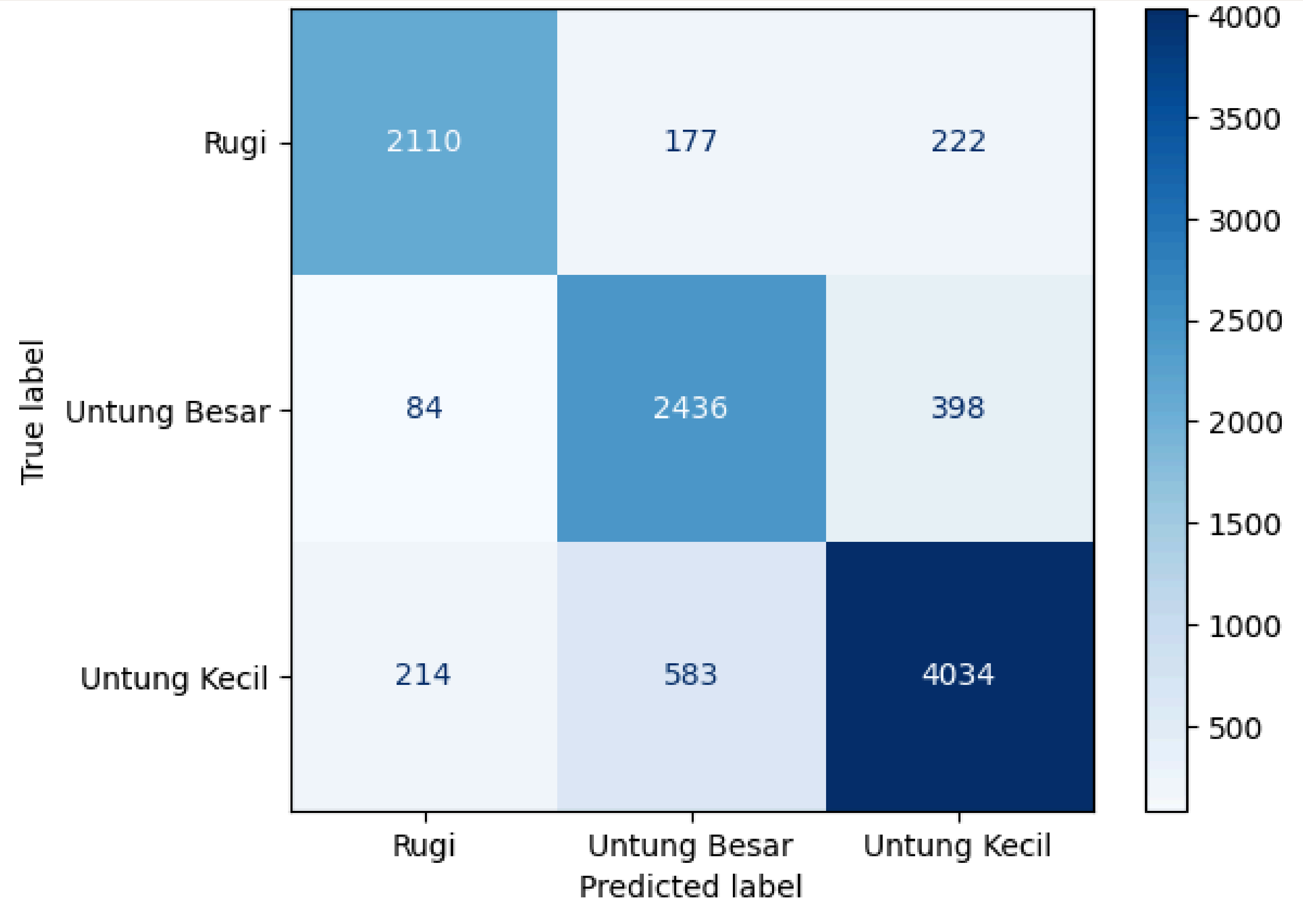
FEATURE

- DISCOUNT
- SALES
- CATEGORY
- MARKET
- SUB_CATEGORY
- SHIPPING_COST
- PRODUCT_NAME
- COUNTRY
- REGION
- QUANTITY
- SHIP_DATE
- CUSTOMER_NAME

Model Accuracy

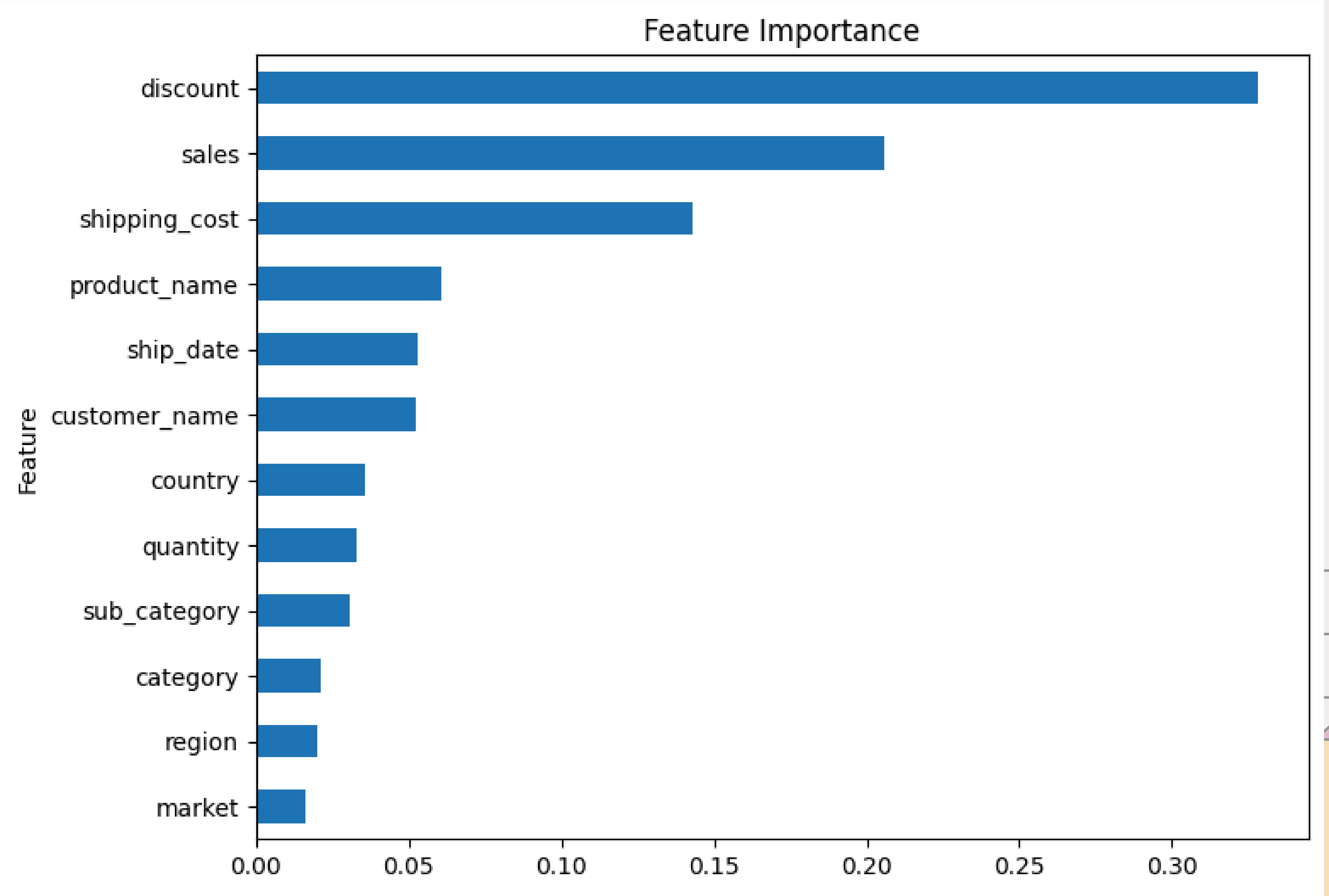
Model	Accuracy
Random Forest	**84%**
XGBoost + Optuna	84%
XGBoost	83%
StackingClassifier	83.4%
BaggingClassifier	82%
GradientBoostingClassifier	82%
VotingClassifier	77%
KNN	58%

Random Forest



ACCURACY 84%
ROC AUC SCORE (OVR) 0.9562

Feature Importance by Random Forest



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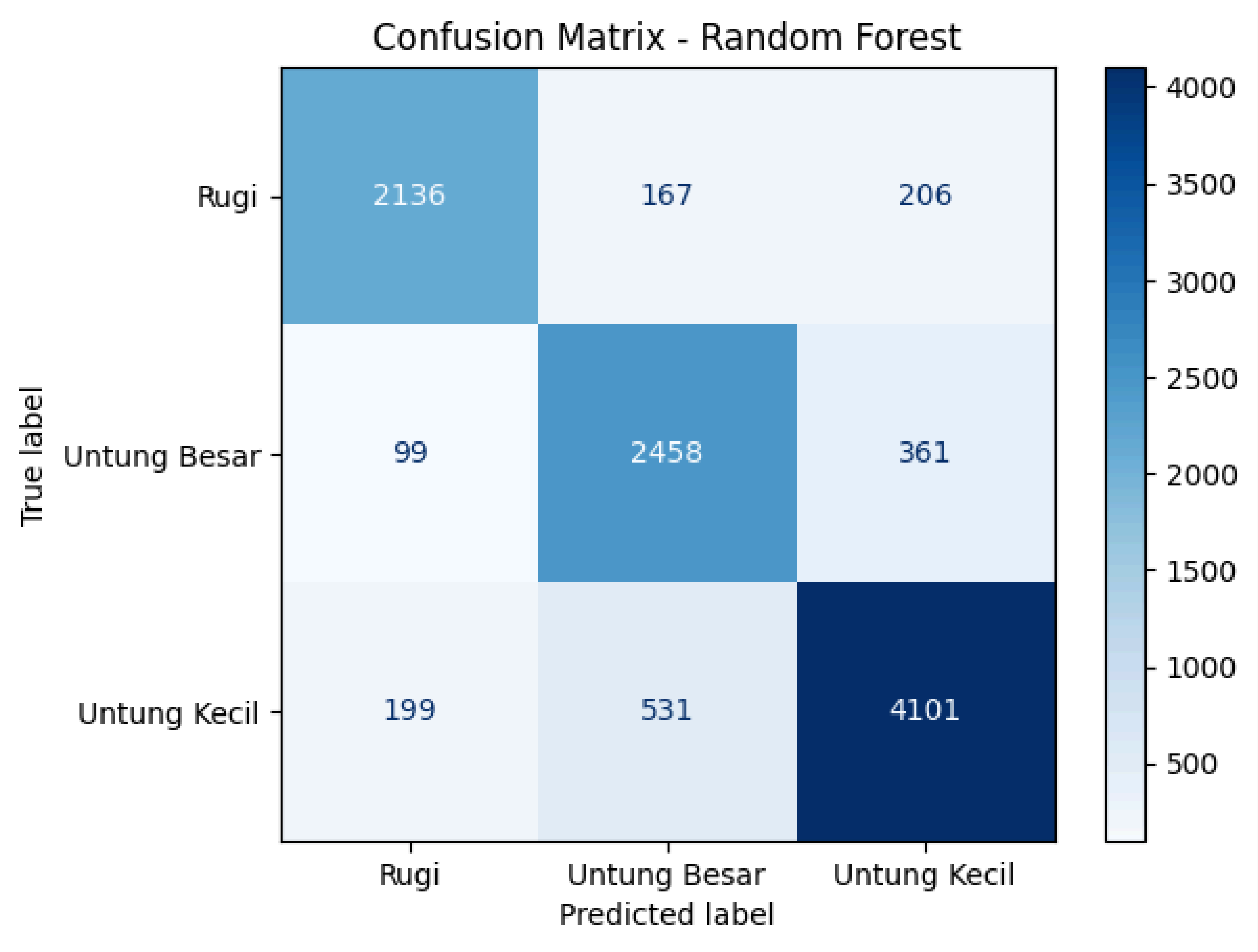
market	1397.6546
order_priority	4.2271
ship_mode	3.1470
segment	1.0461

Model dengan 6 Feature

FEATURE

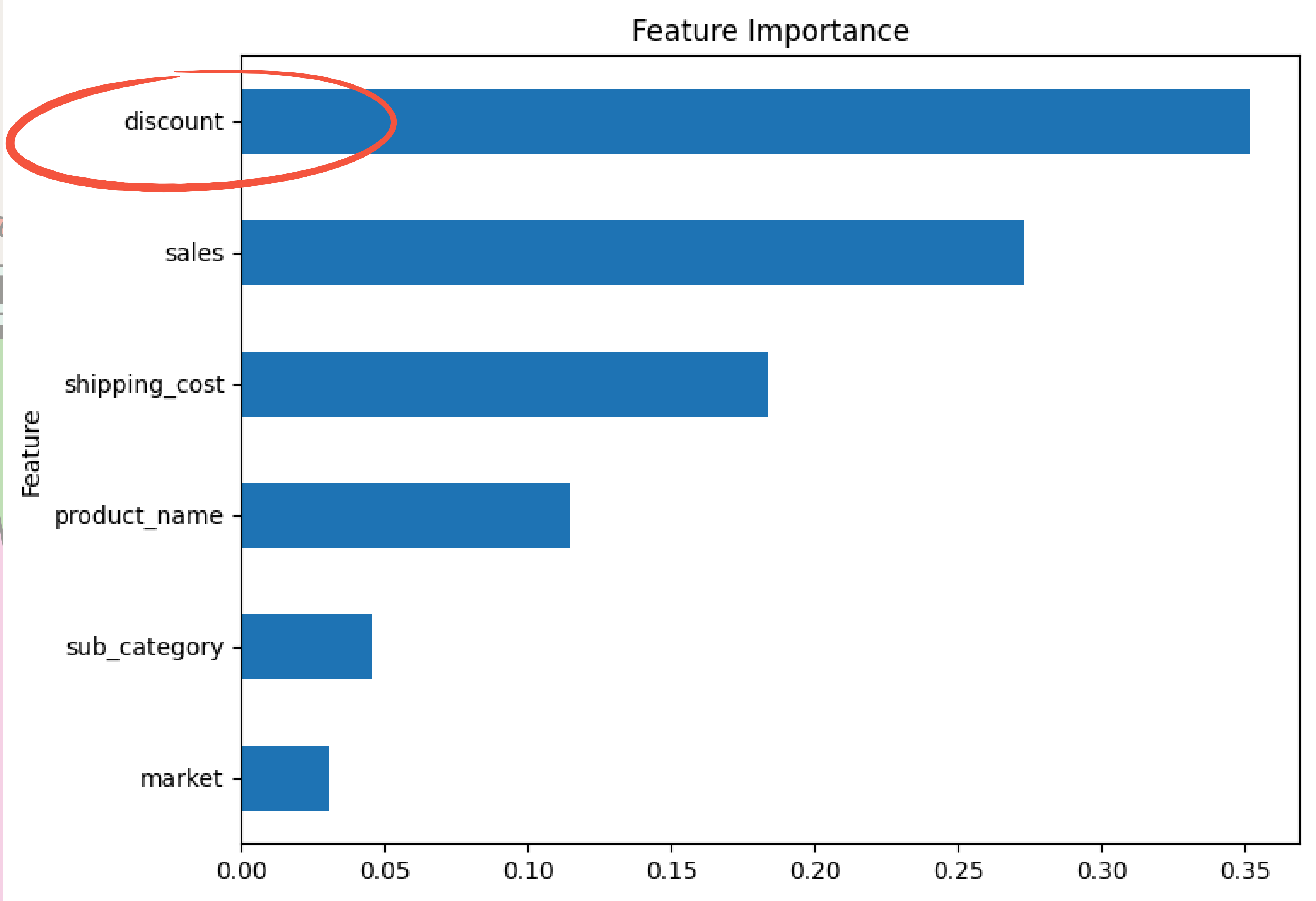
- DISCOUNT
- SALES
- MARKET
- SUB_CATEGORY
- SHIPPING_COST
- PRODUCT_NAME
- ~~COUNTRY~~
- ~~REGION~~
- ~~QUANTITY~~
- ~~SHIP_DATE~~
- ~~CUSTOMER_NAME~~
- ~~CATEGORY~~



ACCURACY 85%

ROC AUC SCORE (OVR) 0.9616

Model dengan 6 Feature



=== Feature Importance ===

	Feature	RF Importance
0	discount	0.351595
1	sales	0.272951
2	shipping_cost	0.184113
3	product_name	0.115046
4	sub_category	0.045771
5	market	0.030524

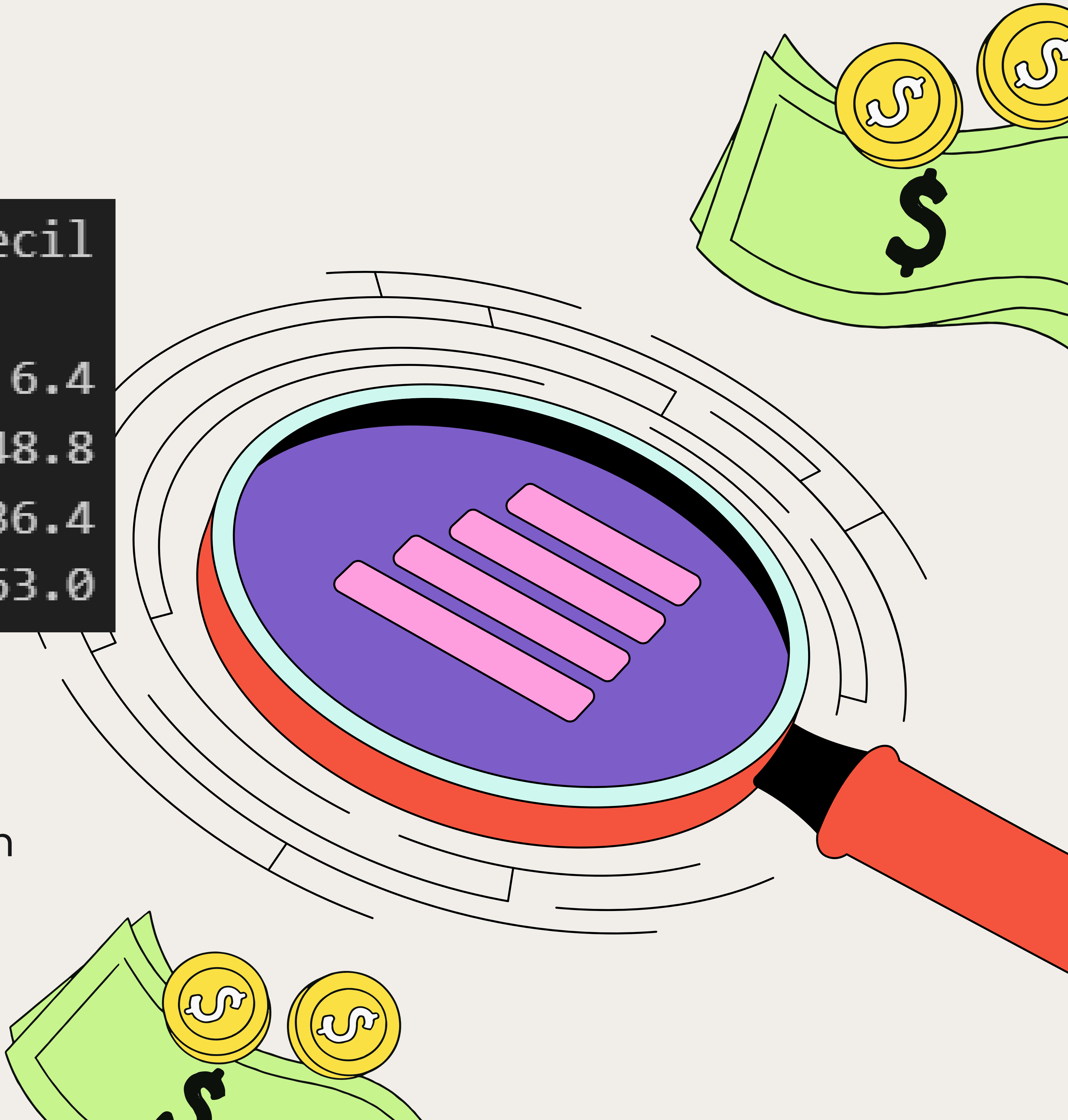
Diskon masih menjadi variabel utama yang mempengaruhi profit

Analisis Lanjutan

profit_bin discount_bin	Rugi	Untung Besar	Untung Kecil
3	92.5	1.2	6.4
2	28.5	22.7	48.8
1	19.2	44.4	36.4
0	0.0	37.0	63.0

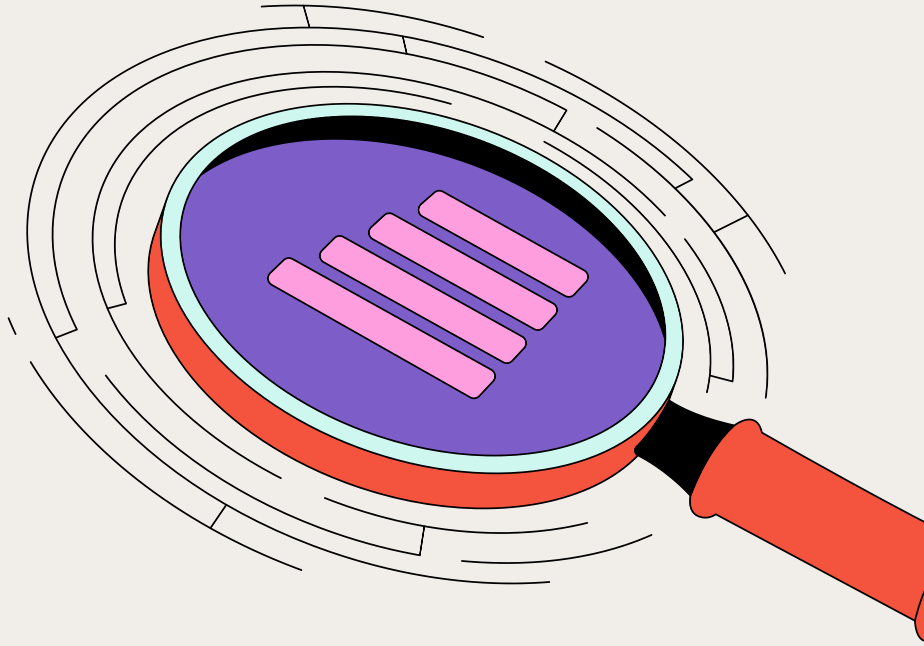
92.5% transaksi dengan diskon di atas 30% berakhir **Rugi!** Hanya 1.2% yang Untung Besar, dan 6.4% Untung Kecil.

✅ Kesimpulan: Diskon terlalu tinggi justru menyebabkan kerugian besar. Ini sinyal bahaya untuk strategi pricing.



PERLU PENDEKATAN BARU

Optimalisasi Besar
Diskon



TANPA MENGORBANKAN MARGIN

Maksimalkan diskon
kurang dari 30%

Kesimpulan

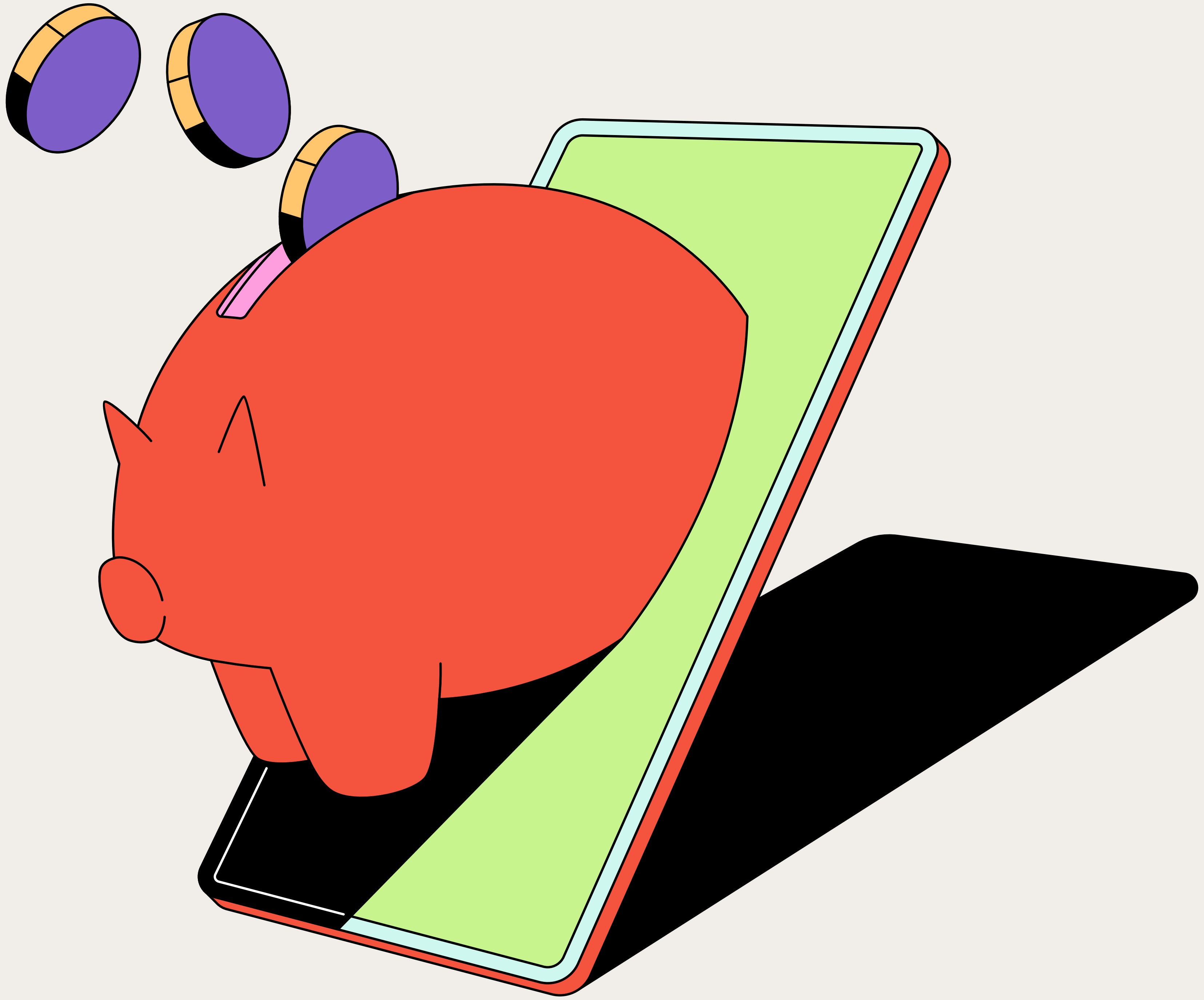
Accuracy model prediksi profit 85%

Diskon tinggi → justru meningkatkan beban rugi.

Evaluasi kembali pengaruh diskon terhadap sales

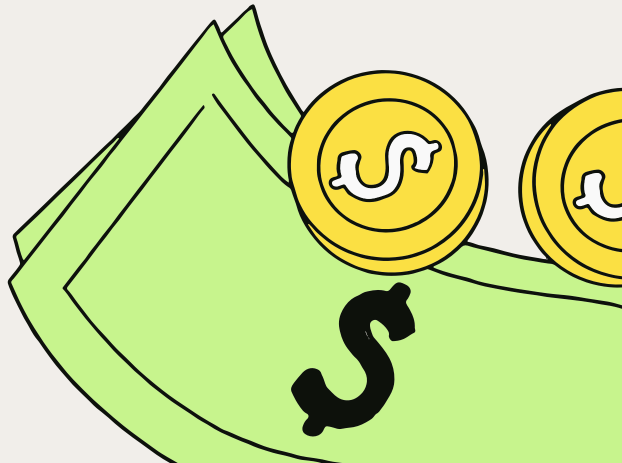
By Data, diskon kurang dari 30% menghasilkan profit yang cukup besar

Ide Tambahan

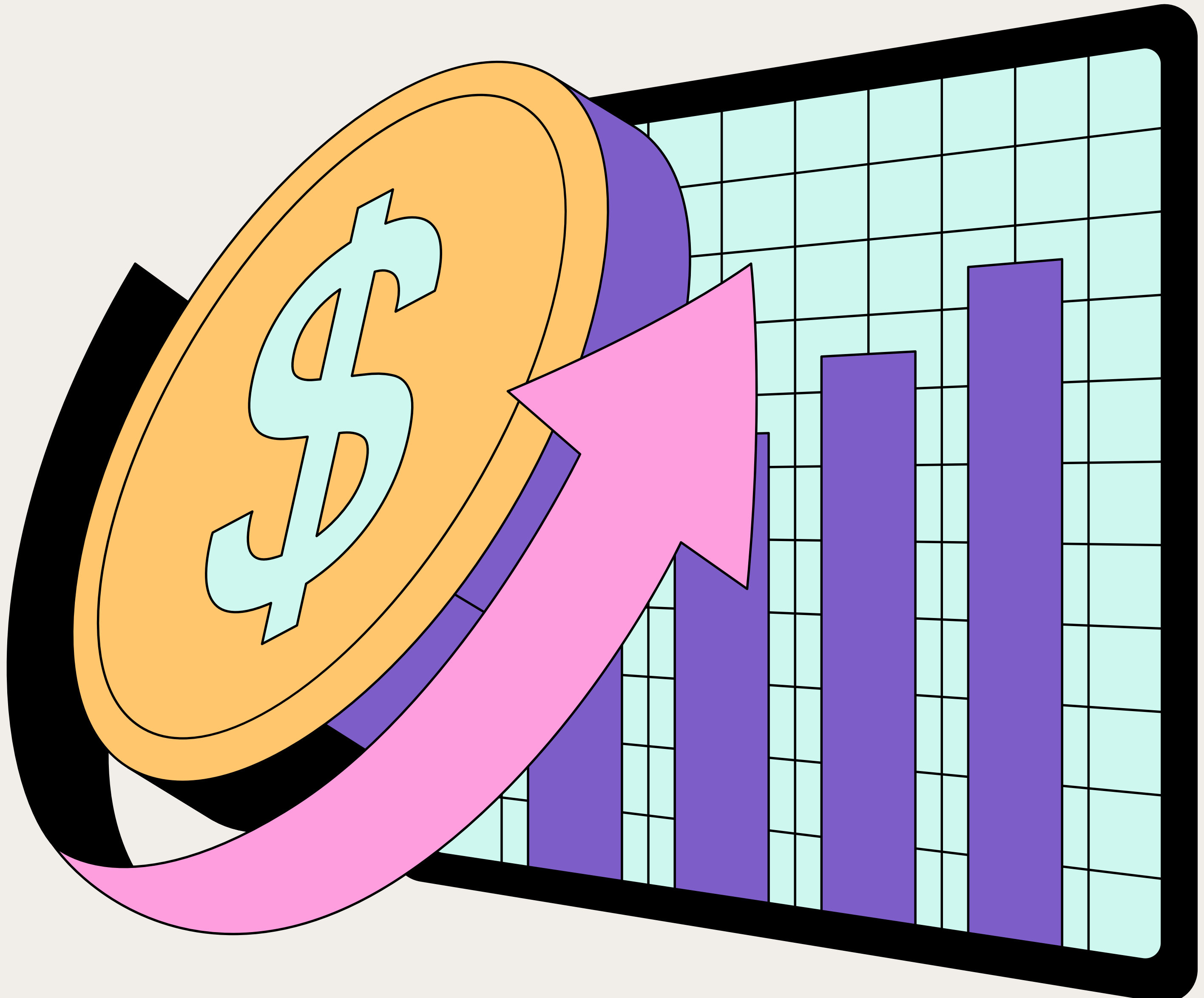


Penerapan SLA (Service Level Agreement) untuk pengiriman

Customer akan mendapat voucher discount hanya apabila mengalami keterlambatan pengiriman

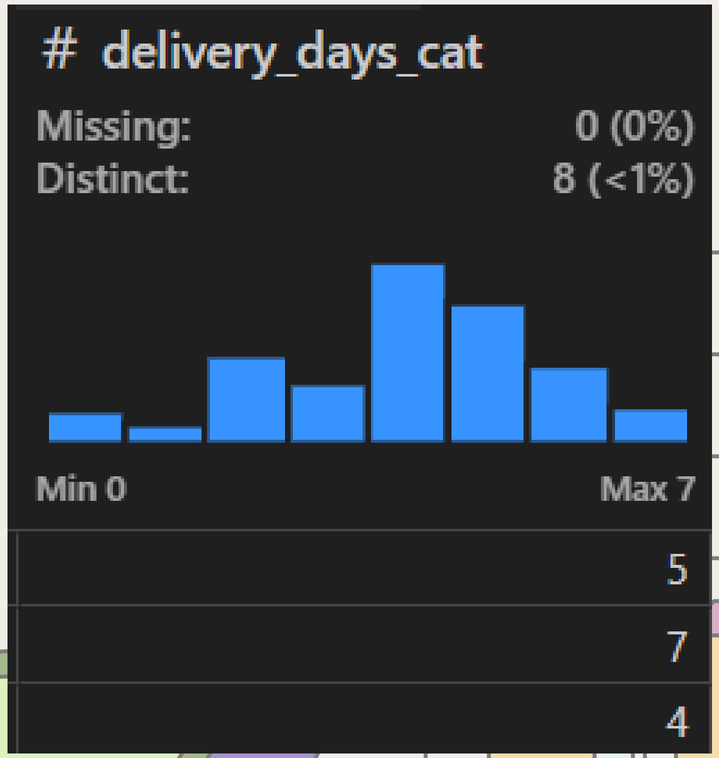


Benefit



- Cost Effective
- Menciptakan Repeat Order
- Meningkatkan Loyalitas Pelanggan

Membangun Model Delivery Speed

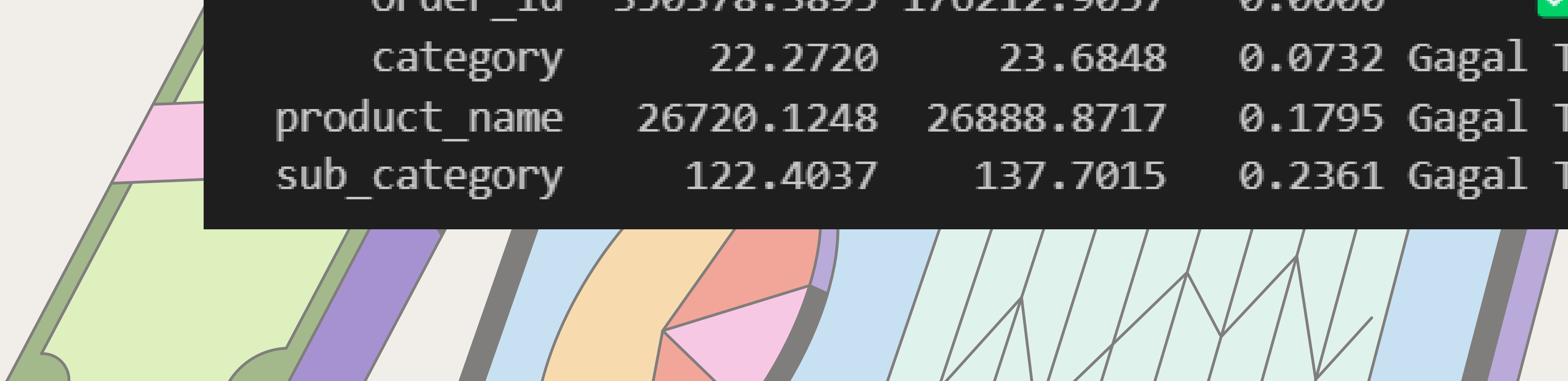


delivery_days_cat	
4 hari	14434
5 hari	11221
2 hari	7026
6 hari	6255
3 hari	5035
7 hari	3057
0 hari	2600
1 hari	1662

Chi Squared Test Of Independence

======= Hasil Uji Chi-Squared terhadap 'Delivery Speed' =======

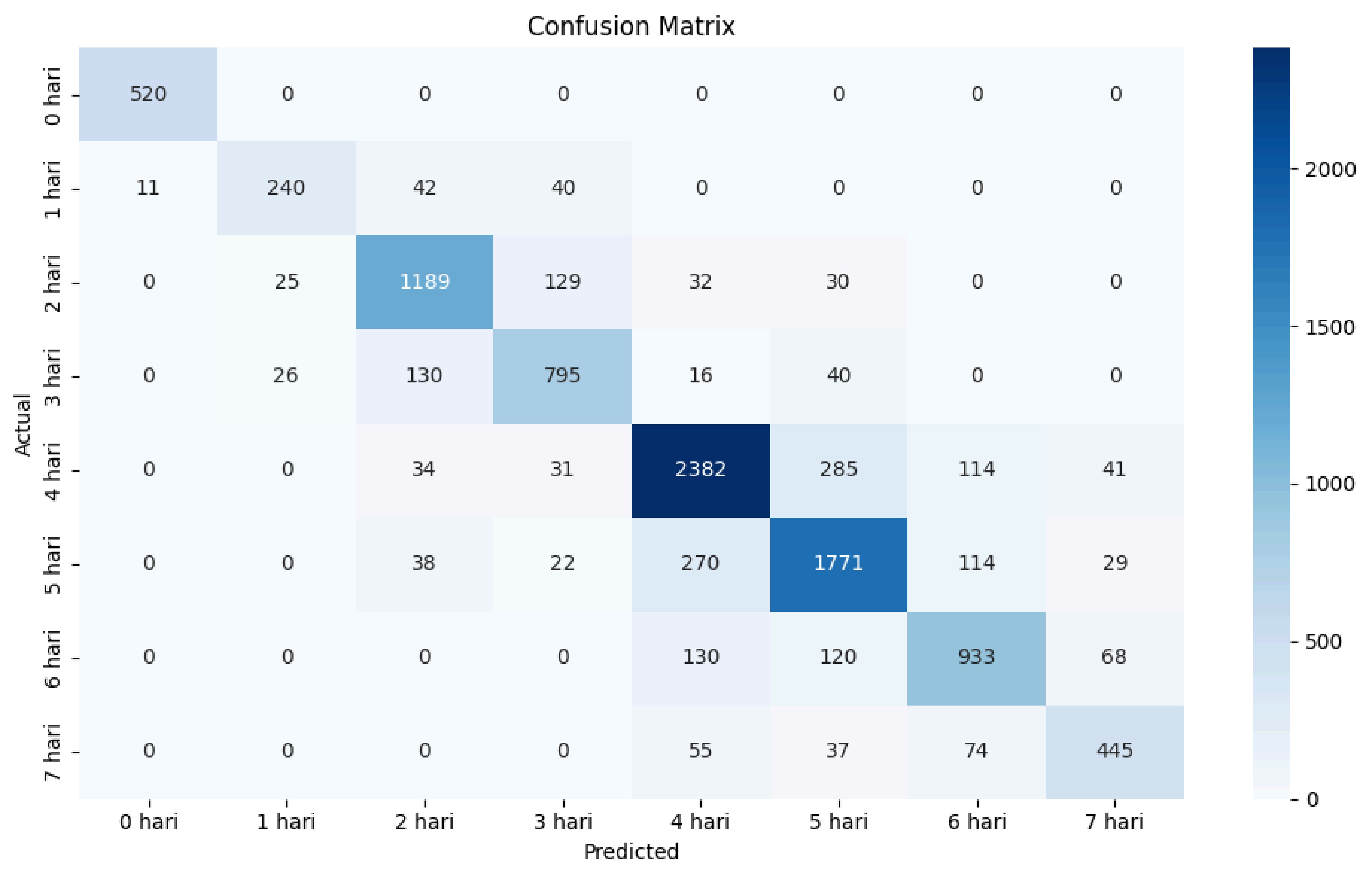
Feature	Chi2 Hitung	Chi2 Tabel	p-value	Kesimpulan
segment	44.9272	23.6848	0.0000	✅ Tolak H0
region	319.3392	106.3948	0.0000	✅ Tolak H0
ship_mode	93450.7180	32.6706	0.0000	✅ Tolak H0
order_priority	30136.6393	32.6706	0.0000	✅ Tolak H0
country	2798.0755	1097.4844	0.0000	✅ Tolak H0
ship_date	38617.3511	10477.5353	0.0000	✅ Tolak H0
market	153.1353	58.1240	0.0000	✅ Tolak H0
customer_name	16127.6794	5732.5505	0.0000	✅ Tolak H0
order_id	350378.3895	176212.9057	0.0000	✅ Tolak H0
category	22.2720	23.6848	0.0732	Gagal Tolak H0
product_name	26720.1248	26888.8717	0.1795	Gagal Tolak H0
sub_category	122.4037	137.7015	0.2361	Gagal Tolak H0



Model Prediction

Accuracy: 81%

ROC AUC Score: 0.9665



Feature Importance

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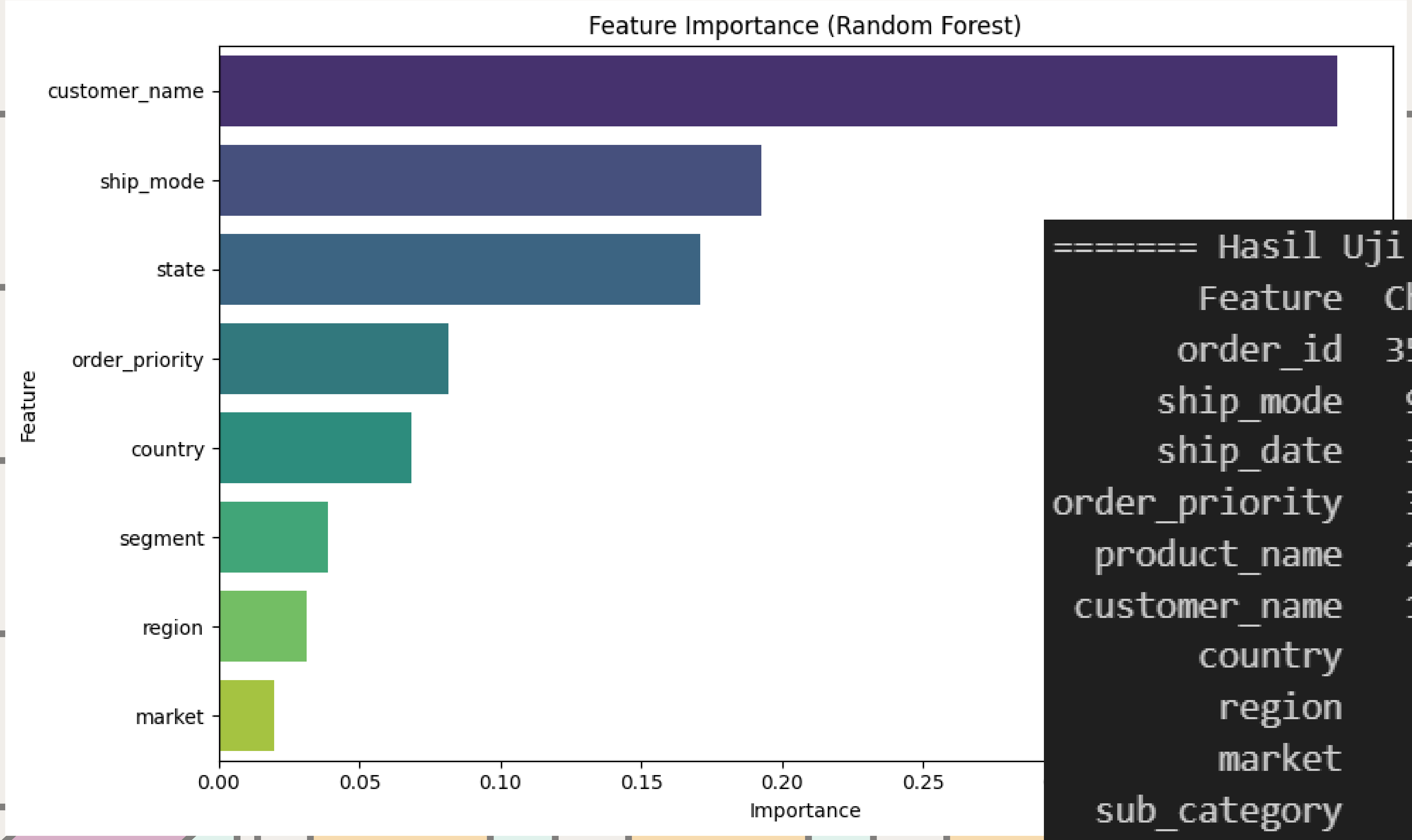

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Kesimpulan

Accuracy model prediksi profit 85%

Accuracy model prediksi delivery speed 81%

Diskon tinggi ≠ loyalitas → justru meningkatkan beban rugi.

Penerapan voucher berbasis SLA memberikan win-win solution: biaya lebih terkendali, pelanggan tetap puas.

Action Plan

Evaluasi Kembali Diskon dan Model Machine Learning

PIC	Langkah	Deskripsi
Data/IT	Integrasi model prediksi profit & delivery	Gabungkan ke sistem operasional/sales pipeline
Marketing	Uji coba strategi SLA-voucher	Terapkan ke sebagian pelanggan high-priority
Finance & CS	Monitoring hasil	Evaluasi dampak ke margin dan retensi pelanggan
Management	Roll-out penuh	Ekspansi strategi ke semua area

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